Module 8
Data Reduction

Measurement Level	Assumption	Possible Techniques
Continuous	Small sample	Metric Multidimensional Scaling
	Assumptions of normality or linearity are not met	Metric Multidimensional Scaling
		Robust Principal Component Analysis
		Robust Exploratory Factor Analysis
		Bayesian Methods
	Assumptions are met	Principal Component Analysis (PCA)
		Exploratory Factor Analysis (EFA)
Ordinal	Small sample	
	Assumptions of normality or linearity are not met	Robust PCA with Polychoric Correlations
		EFA with Polychoric Correlations
		Nonmetric Multidimensional Scaling
		Categorical PCA (CATPCA)
		Multiple Correspondence Analysis
	Assumptions are met	PCA with Polychoric Correlations
		EFA with Polychoric Correlations
2 / 3+ categories	Small sample	
	Assumptions of normality or linearity are not met	PCA with Tetrachoric Correlations
		EFA with Tetrachoric Correlations
		Nonmetric Multidimensional Scaling
		Categorical PCA (CATPCA)
		Multiple Correspondence Analysis
	Assumptions are met	PCA with Tetrachoric Correlations
		EFA with Tetrachoric Correlations